



Lab 6: Data Link Layer Flow Control and ARQ Protocols

Introduction to Data Link Layer Flow Control and ARQ

In computer networks, reliable data transmission is essential for ensuring that information is delivered accurately between devices. The Data Link Layer, which is the second layer in the OSI model, is responsible for node-to-node data transfer and handles error detection and correction. Two critical functions performed at this layer are flow control and error control through Automatic Repeat Request (ARQ) protocols.

Flow Control is the technique used to regulate the amount of data that a sender can transmit before receiving acknowledgments from the receiver. This prevents overwhelming a slower receiver with data it cannot process.

Automatic Repeat Request (ARQ) protocols are error-control mechanisms that use acknowledgments and timeouts to achieve reliable data transmission over unreliable services. When errors are detected, these protocols request retransmission of corrupted or lost frames.

Three widely used ARQ protocols that we will explore in this lab are:

- **Stop-and-Wait ARQ:** A simple protocol where the sender transmits a single frame and waits for its acknowledgment before sending the next one.
- **Go-Back-N ARQ:** A sliding window protocol that allows the sender to transmit multiple frames before requiring an acknowledgment, but if an error occurs, all subsequent frames must be retransmitted.
- **Selective Repeat:** A mechanism that controls the amount of unacknowledged data a sender can have in transit, improving efficiency by adjusting the window size based on network conditions.

Stop-and-Wait ARQ Protocol Simulation

In Stop-and-Wait ARQ, the sender transmits a frame and then waits for an acknowledgment (ACK) from the receiver before sending the next frame. If an ACK is not received within a specified timeout period, the sender assumes the frame was lost or corrupted and retransmits it.

The process works as follows:

1. The sender transmits a frame with a sequence number.
2. The sender starts a timer and waits for an ACK.
3. If the receiver successfully receives the frame, it sends an ACK with the same sequence number.
4. If the sender receives the ACK before the timer expires, it sends the next frame.
5. If the timer expires before an ACK is received, the sender retransmits the same frame.

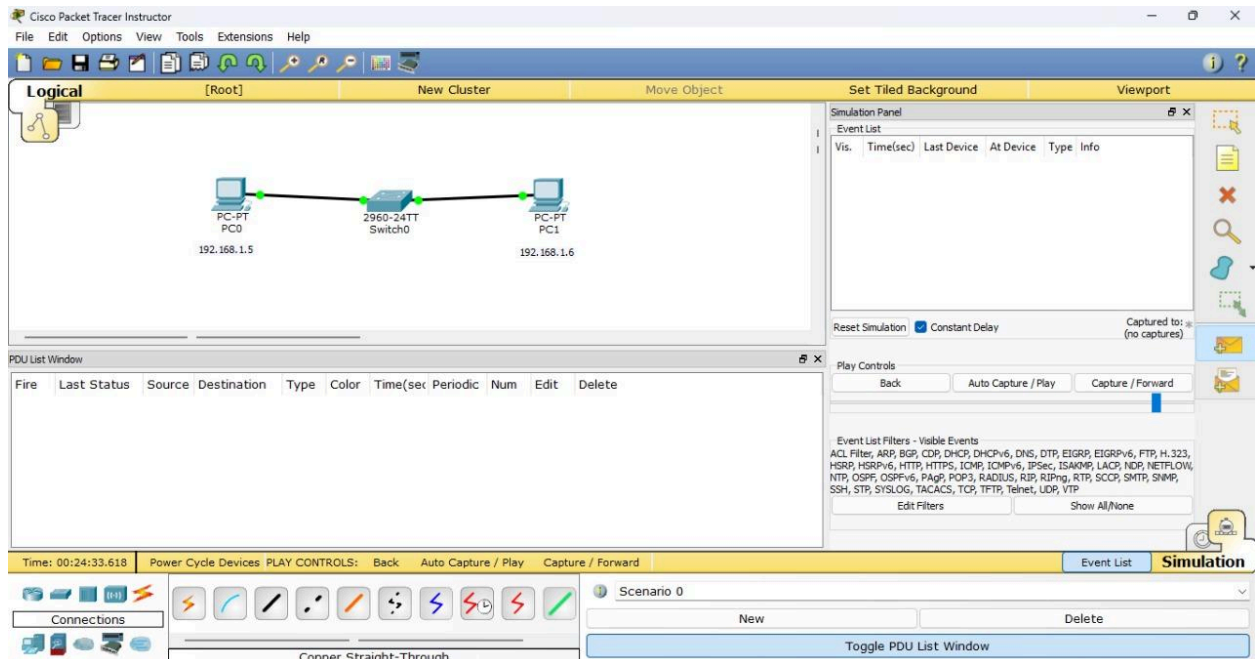


Simulating Stop-and-Wait ARQ Protocol in CISCO Packet Tracer



Setting up the network

To simulate Stop-and-Wait ARQ in Cisco Packet Tracer, start by opening CISCO and creating a simple network setup. Drag and drop two PCs and one switch onto the workspace. Use copper straight-through cables to connect PC1 and PC2 to the switch. Assign IP addresses to both devices by clicking on PC1, navigating to the desktop tab, selecting IP configuration, and setting the IP address to 192.168.1.5 with a subnet mask of 255.255.255.0. Repeat the process for PC2, assigning it the IP address 192.168.1.6 with the same subnet mask. This ensures that both devices are on the same network and can communicate with each other.



PDU List Window Toggled



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Once the network is set up, switch to simulation mode by clicking the clock icon at the bottom right of the screen. In this mode, network packets and their transmission can be observed in detail. To simulate packet transmission using Stop-and-Wait ARQ, select the "Toggle PDU Tool List" from the toolbar and click on PC1 followed by PC2. This action sends a packet from PC1 to PC2, and in Stop-and-Wait ARQ, PC1 will wait for an acknowledgment before sending the next packet. The event list at the bottom will display the transmission and acknowledgment process.

The screenshot displays the Cisco Packet Tracer Instructor interface. The main workspace shows a network topology with two PCs (PC0 and PC1) connected via a 2960-24TT switch. PC0 has IP 192.168.1.5 and PC1 has IP 192.168.1.6. The interface includes a menu bar (File, Edit, Options, View, Tools, Extensions, Help), a toolbar, and several panels.

Simulation Panel: Shows the Event List with the following data:

Vis.	Time(sec)	Last Device	At Device	Type	Info
	0.000	--	PC0	ICMP	
	0.000	--	PC0	ARP	

PDU List Window: Shows the following data:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	In Progress	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

The bottom status bar shows the time as 00:24:33.618 and the simulation mode as Simulation. The toolbar includes icons for connections, power cycle, devices, and simulation controls.

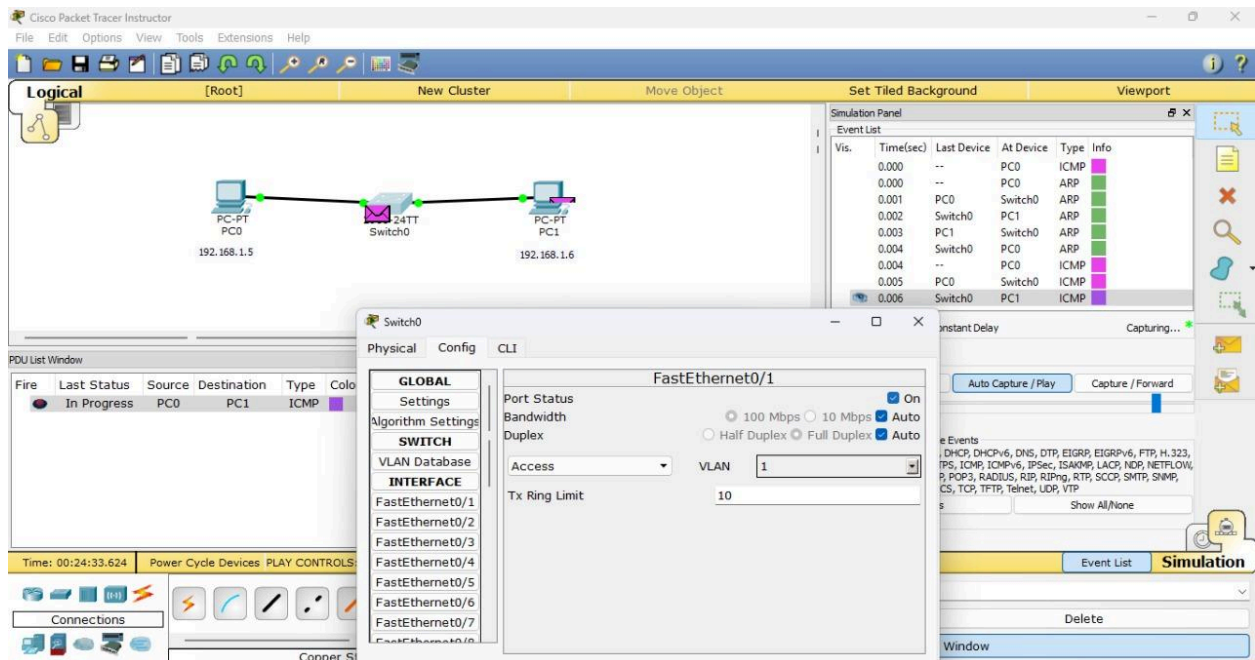
Sending packets and viewing the event list



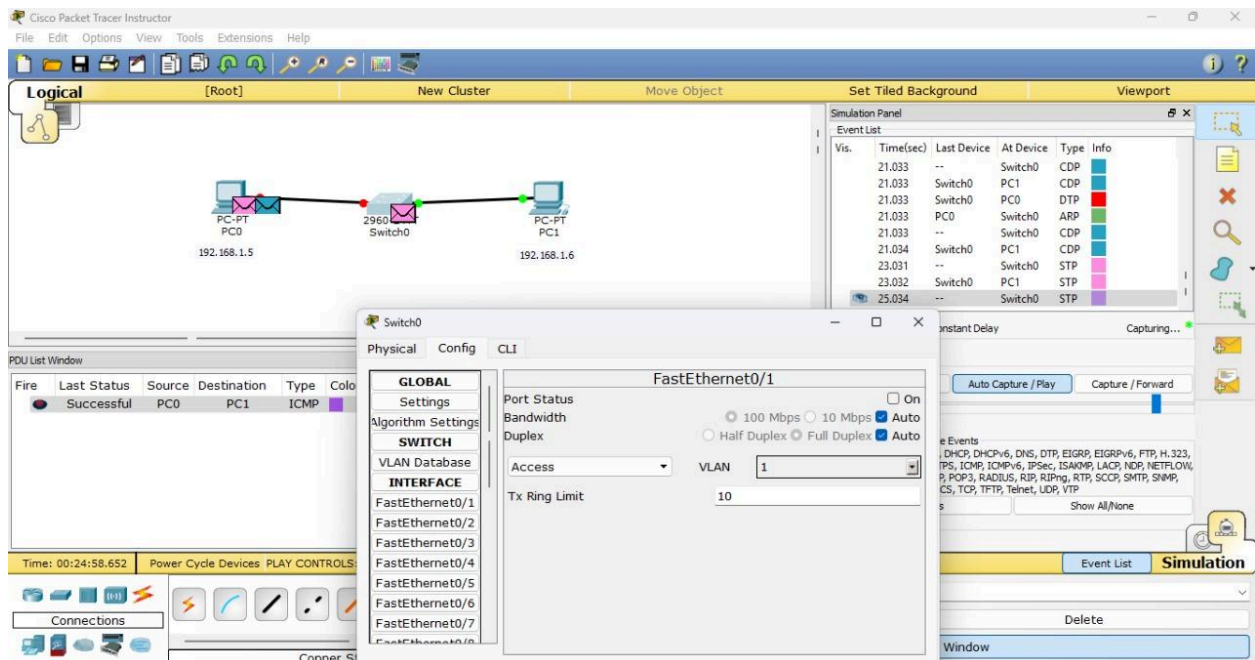
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Sending packets with the port status "on"



Turning port status "off"

To introduce packet loss and observe retransmission, one method is to manually toggle the switch port off and on. Click on the switch, go to the "Config" tab, and under "Port Settings," select the port connected to PC1, such as FastEthernet 0/1. Turn the port status off, wait for a



few seconds, and turn it back on. This disruption simulates a lost packet because the acknowledgment will not be received, triggering a retransmission from PC1. The retransmission attempt can be seen in the event list, confirming that Stop-and-Wait ARQ is functioning correctly by ensuring successful delivery of each packet.

Note: An alternative method to simulate packet loss is by using Cisco CLI commands on the switch. Open the switch CLI and enter configuration mode by typing `enable`, followed by `configure terminal`. Identify the interface connected to PC1, such as interface `FastEthernet 0/1`, and shut it down using the `shutdown` command. This action effectively drops packets by disabling the port. After a brief delay, reactivate the port with the `no shutdown` command. This method achieves the same effect as manually toggling the port in the GUI but allows for more precise control over when and how the packet loss occurs.

After introducing packet loss, observe how PC1 retransmits the lost packet after a timeout. In Stop-and-Wait ARQ, PC1 will not proceed with new data transmission until it successfully receives an acknowledgment for the lost packet.

By the end of this simulation, it becomes evident how Stop-and-Wait ARQ handles errors. This method ensures reliable delivery but at the cost of efficiency, as only one packet is transmitted at a time, leading to delays in high-latency networks. The advantages include simplicity and guaranteed packet delivery, while the major drawback is the waiting time after each transmission, reducing overall throughput. This trade-off highlights why other ARQ methods, such as Go-Back-N, are preferred for high-speed data transfer scenarios.

Go-Back-N ARQ Protocol Simulation

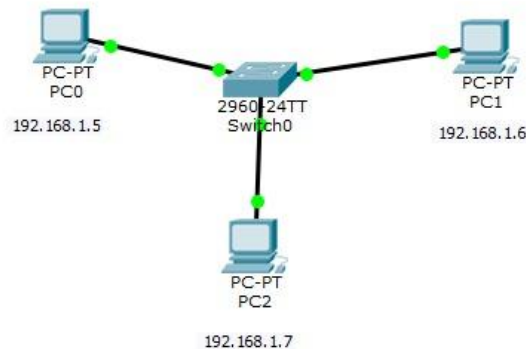
Go-Back-N ARQ is a more efficient protocol that allows the sender to have multiple frames in transit before receiving acknowledgments. It uses a sliding window of size N to control the number of unacknowledged frames.

The protocol works as follows:

1. The sender can transmit up to N frames without receiving an acknowledgment.
2. Each frame is assigned a sequence number within a range (typically 0 to N-1).
3. The receiver acknowledges correctly received frames by sending an ACK with the sequence number of the next expected frame.
4. If an error is detected in frame n, the receiver discards frame n and all subsequent frames.
5. When the sender receives a duplicate ACK or experiences a timeout, it retransmits all frames starting from the unacknowledged frame.

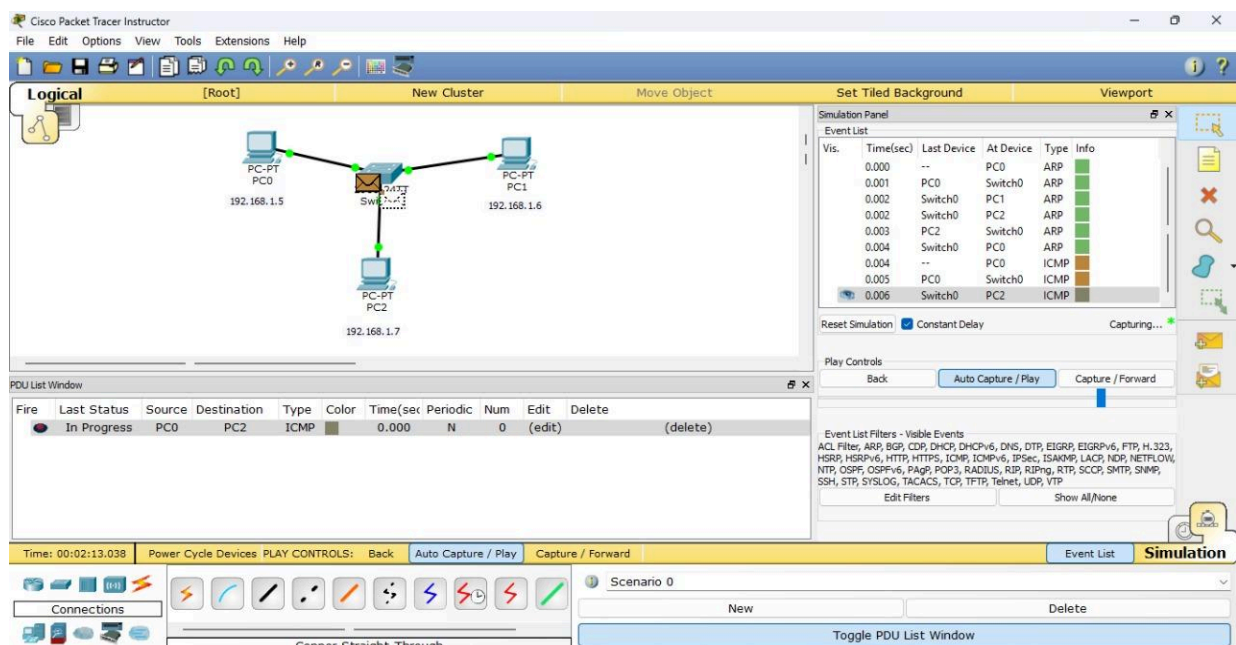


Simulating Go-Back-N ARQ Protocol in CISCO Packet Tracer



Setting up the network

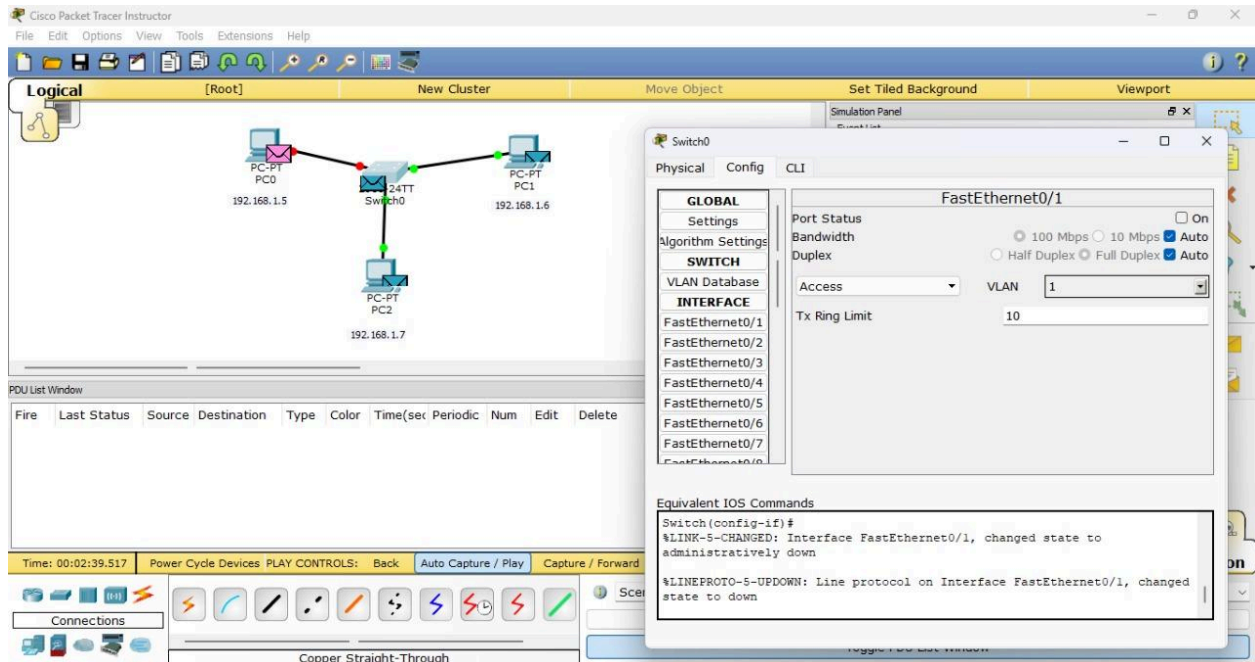
To simulate Go-Back-N ARQ in Cisco Packet Tracer, start by creating a network that consists of three PCs and a switch. Open Packet Tracer, place PC1, PC2, and PC3 in the workspace, and connect them using copper straight-through cables to a central switch. Assign IP addresses to each device by clicking on PC1, navigating to the desktop tab, selecting IP configuration, and setting the IP address to 192.168.1.5 with a subnet mask of 255.255.255.0. Repeat this for PC2 with an IP address of 192.168.1.6 and PC3 with an IP address of 192.168.1.7, ensuring they all belong to the same network.



Sending packets from PC1 to PC 3



Once the network is set up, switch to simulation mode by selecting the clock icon at the bottom right. In this mode, individual packets can be observed as they travel across the network. To begin transmitting packets from PC1 to PC3, use the "Toggle PDU List Window" and click on PC1 followed by PC3. This simulates packet flow, but since Packet Tracer does not have a built-in option to specify the exact number of packets that can be sent before a retransmission, Go-Back-N behavior is inferred based on packet loss and subsequent retransmissions.



Retransmission of multiple packets after turning the port status "off"

To introduce errors and observe the behavior of Go-Back-N ARQ, manually toggle the switch port off and on. Click on the switch, go to the "Config" tab, and under "Port Settings," locate the port connected to PC2 (such as FastEthernet 0/2) and turn the port status off. This disrupts communication between PC1 and PC3, leading to multiple packet losses. After a few seconds, turn the port back on to restore connectivity. Unlike Stop-and-Wait ARQ, where only one lost packet is retransmitted at a time, Go-Back-N requires retransmitting all packets from the last successfully acknowledged packet onward. Since Packet Tracer does not allow explicit control over the window size, this behavior is observed through repeated transmissions after an interruption.

To further verify this behavior, monitor the event list and track how packets are being sent and resent. When a packet is lost due to the switch port being turned off, PC1 will eventually resend not just the lost packet but a series of subsequent packets as well, illustrating the Go-Back-N approach. This happens because, in Go-Back-N ARQ, if an error is detected in a single packet,



all subsequent packets in the window must be retransmitted.

Although Packet Tracer does not provide an explicit way to set the "N" value or window size manually, the simulation effectively demonstrates Go-Back-N behavior by showing how multiple packets are retransmitted in response to a single loss. This approach improves efficiency over Stop-and-Wait ARQ by allowing multiple packets to be in transit before receiving acknowledgments, but it has the drawback of redundant retransmissions in case of errors. If a single packet is lost or corrupted, all subsequent packets in the window must be resent, even if they were received correctly the first time. This trade-off highlights the balance between transmission efficiency and error recovery in ARQ protocols.

Selective Repeat Simulation

The Selective Repeat is a flow control mechanism that allows a sender to transmit multiple packets before requiring an acknowledgment. By adjusting the window size, we can optimize network performance based on bandwidth, delay, and error rates.

The Selective Repeat works as follows:

1. The sender maintains a window of size W , which represents the maximum number of unacknowledged frames that can be in transit at any time.
2. Each frame is assigned a unique sequence number within a range (typically 0 to $2^n - 1$, where n is the number of bits used for sequence numbering).
3. The sender's window slides forward as acknowledgments are received, allowing new frames to be transmitted.
4. The receiver maintains its own window of size W' (which may differ from the sender's window size), representing the range of sequence numbers it is willing to accept.
5. The receiver acknowledges frames by sending an ACK containing the sequence number of the next frame it expects to receive.
6. Unlike Go-Back-N, the receiver may selectively acknowledge frames, meaning it can accept and buffer frames that arrive out of order.
7. When the sender detects packet loss (through timeout or duplicate ACKs), it has two options depending on the specific implementation:
 - Selective Repeat: Retransmit only the specific lost frames
 - Cumulative ACK: Retransmit all unacknowledged frames starting from the first lost frame
8. The window size can be dynamically adjusted based on network conditions to optimize throughput and minimize congestion.
9. The protocol uses the window size to implement flow control, preventing the sender from overwhelming the receiver or the network.



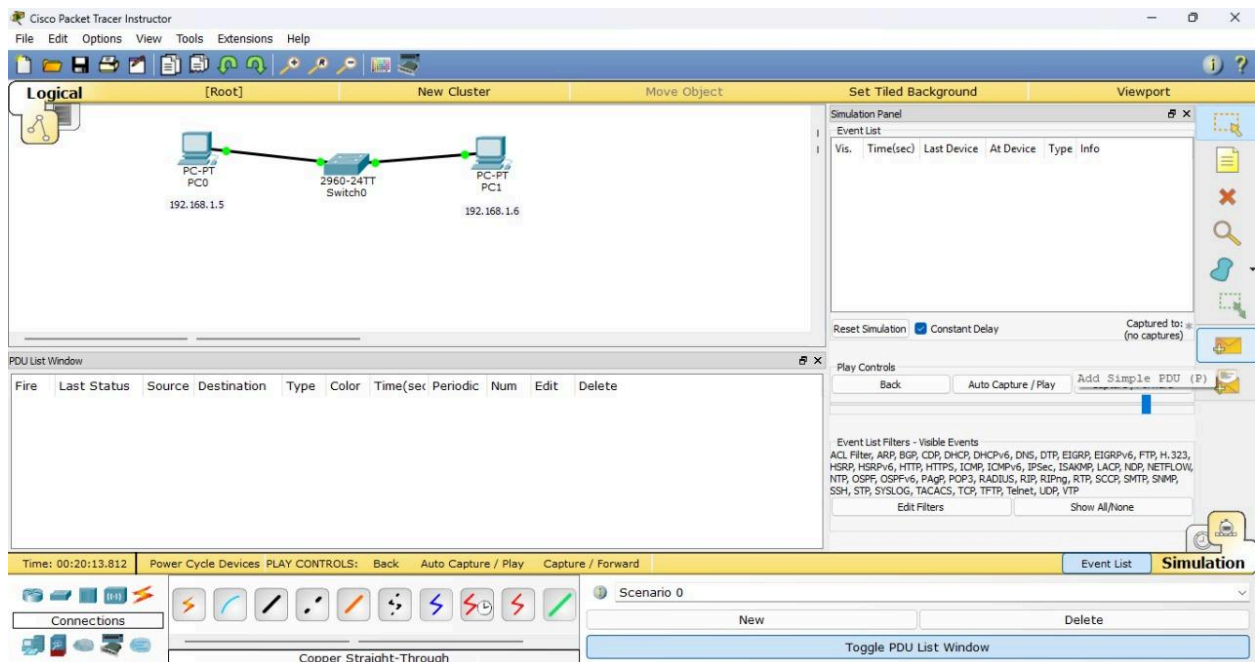
10. When the sender reaches the maximum sequence number, it wraps around and reuses sequence numbers, which is why the window size must be less than half the sequence number space to avoid ambiguity.

Simulating Selective Repeat in CISCO Packet Tracer



Setting up the network

To set up the network, create a simple topology with two PCs (PC1 and PC2) connected through a switch. Assign IP addresses as follows: PC1 (192.168.1.1) and PC2 (192.168.1.2). This setup will allow communication between the two devices.



Selecting "Simple PDU"

In Packet Tracer's Simulation Mode, instead of using ping, use the "Add Simple PDU" option to send multiple packets simultaneously. Click on PC1, then select the envelope icon (Add Simple



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PDU) and click on PC2.

The screenshot shows the Cisco Packet Tracer interface. The network topology consists of PC0 (192.168.1.5) connected to Switch0 (2960-24TT) which is connected to PC1 (192.168.1.6). The PDU List Window shows three ICMP packets in progress from PC0 to PC1. The Simulation Panel shows the Event List with three ICMP packets captured.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	In Progress	PC0	PC1	ICMP	Green	0.000	N	0	(edit)	(delete)
	In Progress	PC0	PC1	ICMP	Yellow	0.000	N	1	(edit)	(delete)
	In Progress	PC0	PC1	ICMP	Blue	0.000	N	2	(edit)	(delete)

Sending multiple packets simultaneously

Repeat this process three times quickly to simulate a burst of packet transmissions.

The screenshot shows the Cisco Packet Tracer interface after sending a burst of packets. The network topology remains the same. The PDU List Window shows three ICMP packets as 'Successful' from PC0 to PC1. The Simulation Panel shows the Event List with six ICMP packets captured, indicating a burst of transmissions.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP	Green	0.000	N	0	(edit)	(delete)
	Successful	PC0	PC1	ICMP	Yellow	0.000	N	1	(edit)	(delete)
	In Progress	PC0	PC1	ICMP	Blue	0.000	N	2	(edit)	(delete)

Sending packets



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Time: 00:21:29.968 Power Cycle Devices PLAY CONTROLS: Back Auto Capture / Play Capture /

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Del
●	Successful	PC0	PC1	ICMP	Green	0.000	N	0	(edit)	
●	Successful	PC0	PC1	ICMP	Green	0.000	N	1	(edit)	
●	In Progress	PC0	PC1	ICMP	Purple	0.000	N	2	(edit)	

Connections

Copper Straight-Through

Equivalent IOS Commands

```
state to up
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#shutdown
Switch(config-if)#
```

Introducing an error

Time: 00:21:42.248 Power Cycle Devices PLAY CONTROLS: Back Auto Capture / Play Capture /

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Del
●	Successful	PC0	PC1	ICMP	Green	0.000	N	0	(edit)	
●	Successful	PC0	PC1	ICMP	Green	0.000	N	1	(edit)	
●	Failed	PC0	PC1	ICMP	Purple	0.000	N	2	(edit)	

Connections

Copper Straight-Through

Equivalent IOS Commands

```
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

Packets resent

Now, to introduce a packet loss scenario, toggle the switch port connected to PC2 OFF while packets are being sent. This simulates an interrupted transmission, where PC1 continues to



send packets without receiving acknowledgments. Once the switch is turned back ON, observe how all unacknowledged packets are retransmitted together (i.e; in a way, only the lost packets are being resent), mimicking a form of sliding window retransmission.

Limitations of CISCO Packet Tracer

While this approach partially simulates the behavior of the selective repeat protocol, it does not fully replicate how TCP handles windowing. In a true sliding window implementation, only lost packets should be retransmitted instead of all outstanding packets. However, Packet Tracer does not allow fine-tuned control over window size or selective retransmission, which limits our ability to fully implement the protocol.

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Stop-and-Wait ARQ Simulation (5 Marks)

Simulate the Stop-and-Wait ARQ protocol by setting up two PCs connected to a switch. PC0 will have the IP address **192.168.1.1**, and PC1 will have **192.168.1.2**. In Cisco Packet Tracer, place the two PCs and a switch, then connect them using straight-through cables. Assign the specified IP addresses to each PC. Open the Command Prompt on PC0 and send a ping request to PC1. Switch to Simulation Mode to observe the packet flow.

To introduce an error, manually **turn off the connection between PC1 (your screenshot should clearly show which interface you are doing this for) and the switch after the request is sent** and turn it back on after a few seconds. Observe how PC0 handles the missing acknowledgment (ACK) and retransmits the data.

Question: What happens if an acknowledgment (ACK) is lost during Stop-and-Wait ARQ? Explain how the sender handles this situation.

Task 2: Go-Back-N ARQ Simulation (5 Marks)

Simulate the Go-Back-N ARQ protocol by setting up three PCs connected through a switch. PC0 will have the IP address **192.168.2.1**, PC1 will use **192.168.2.2**, and PC2 will have



192.168.2.3. In Cisco Packet Tracer, place three PCs and a switch, then connect all PCs to the switch using straight-through cables. Assign the specified IP addresses to each PC. Open the Command Prompt on PC0 and send continuous ping requests to PC2 while monitoring the transmission in Simulation Mode. To introduce an error, **disable PC2's connection to the switch (your screenshot should clearly show which interface you are doing this for) after the first two packets have been sent** and re-enable it after a few seconds. Observe how multiple frames are retransmitted once the connection is restored, as Go-Back-N requires retransmission of all frames after a lost or corrupted one.

Question: What is the impact of using Go-Back-N ARQ on network efficiency when multiple frames are lost in a row?

Task 3: Selective Repeat Protocol Simulation (5 Marks)

Simulate the Sliding Window Protocol using a switch and two PCs. PC0 will have the IP address **192.168.3.1**, and PC1 will have **192.168.3.2**. In Cisco Packet Tracer, set up a switch and connect both PCs to it using straight-through cables. Assign the specified IP addresses to the PCs and configure continuous ping requests from PC0 to PC1 while running the simulation in Simulation Mode. The sliding window size should be considered as **4**, meaning you have to send **4 packets simultaneously**.

To introduce an error, **disable PC1's Ethernet port right after receiving two packets and re-enable it after a few seconds (your screenshot should clearly show which interface you are doing this for)**. Observe how only the unacknowledged packets are retransmitted, demonstrating how the Sliding Window Protocol optimizes network efficiency by avoiding unnecessary retransmissions.

Question 1: What potential issues could arise if the sender's window size is too large compared to the network capacity, and how might this affect overall performance?

Question 2: How does the fixed sliding window size of 4 in this simulation limit the accuracy of real-world protocol behavior, and what changes could be made to better reflect real network conditions?

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number**.
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**.



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- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1. (for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

